



COMSATS University Islamabad

Department of Computer Science

Course Description Form (CDF)

Course Information

Course Code: AIC372

Course Title: **Knowledge Representation and Reasoning**

Credit Hours: 3 (3, 0)

Lecture Hours/Week: 3

Lab Hours/Week: 0

Pre-Requisites: **CSC262- Artificial Intelligence**

Course Objectives

- To understand the fundamental principles of logic-based Knowledge Representation;
- To model simple application domains in a logic-based language;
- To understand the notion of a reasoning service;
- To discuss the fundamentals of the reasoning algorithms underlying current systems;
- To explain the fundamental trade-off between representation power and computational properties of a logic-based representation language.

Course Content

Representing knowledge using Logic; The Language of First Order Logic: Syntax, Semantics, Explicit and Implicit Belief; Expressing Knowledge: Knowledge Engineering, Vocabulary, Basic & Complex Facts, Terminological Facts, Entailments; Resolution: The Propositional Case, Handling Variables & Quantifiers, Computational Intractability; Reasoning with Horn Clauses; Procedural Control of Reasoning; Rules in Production Systems; Object Oriented Representation; Structured Descriptions; Ontology Languages for the Semantic Web; Non-monotonic logics.

Unit wise Major Topics

Unit	Topic	No. of teaching hours
1.	Knowledge: Overview, Types, and Relation between Knowledge & Intelligence.	3
2.	Knowledge Representations (KR): Cycle of Knowledge Representation in AI, Techniques, Components of KR Logical Representation, Frame, Production Rules, Inheritance & Defaults, Object Oriented Representation, Structured Descriptions, Knowledge Graphs & Bases, & Representing Knowledge in an Uncertain Domain, and Building KR Systems	10.5
3.	Expressing Knowledge: Knowledge Engineering, Vocabulary, Basic & Complex Facts, Terminological Facts, and Entailments.	7.5
4.	First Order Logic (FOL): Syntax & Semantics, Unification, Representation in FOL, Resolution Refutation in FOL, FOL with Equality, Complexity of Theorem Proving; and Explicit & Implicit Belief.	9
5.	Reasoning: Tradeoff between Representation & Reasoning; Symbolic Reasoning; Resolution: The Propositional Case, Handling Variables & Quantifiers; Computational Intractability; Reasoning with Horn Clauses Rules in Production Systems; Probabilistic Reasoning; Fuzzy Reasoning, and Procedural Control of Reasoning.	9

6.	Ontology Languages for the Semantic Web; and Non-monotonic Logics.			6	
Total Contact Hours				45	
Mapping of CLOs and GAs					
Sr.#	Unit #	Course Learning Outcomes	Blooms Taxonomy Learning Level	GA	
CLO's for Theory					
CLO-1	1	Explain types of knowledge and their relation to intelligence.	Understanding	2	
CLO-2	2,3	Build a context-specific knowledge representation system for a given problem.	Creating	3-4	
CLO-3	4	Demonstrate deductive systems using first order logic.	Applying	2,3	
CLO-4	5,6	Develop reasoning-based Intelligent Systems.	Creating	3-5	
CLO Assessment Mechanism					
Assessment Tools		CLO-1	CLO-2	CLO-3	CLO-4
Quizzes		Quiz 1	Quiz 2	Quiz 3	Quiz 4
Assignments		-	Assignment 1 & 2	Assignment 3	Assignment 4
Mid Term Exam		Mid Term Exam	Mid Term Exam	Mid Term Exam	
Final Term Exam		Final Term Exam			
Text and Reference Books					
Textbooks:					
1. Knowledge representation and reasoning, Blokdyk, G., 5STARCooks, 2020.					
Reference Books:					
1. Knowledge Representation, Reasoning, and the Design of Intelligent Agents: The Answer-Set Programming Approach, Gelfond, M., Kahl, Y., Cambridge University Press, 2014.					
2. Knowledge Representation and Reasoning, Brachman, R., Levesque, H., The Morgan Kaufmann Series in Artificial Intelligence, 2004.					